Addis Ababa Science & Technology University

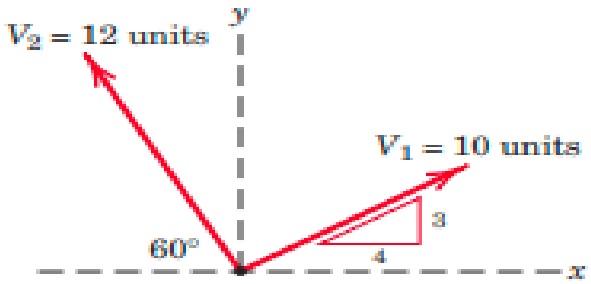
College of Architecture & Civil Engineering

Department of Civil Engineering

Engineering Mechanics (Statics) for Pre-Engineering Students

Work Sheet for Chapter-1

1. Given the following three vectors $\vec{V}_1=2\hat{\imath}+\hat{\jmath}-\hat{k}$ and $\vec{V}_2=4\hat{\imath}-3\hat{\jmath}$ and $\vec{V}_3=-3\hat{\imath}+2\hat{\jmath}-5\hat{k}$.Determine the magnitude and direction of vector $\vec{V}$ given by $\vec{V}=\vec{V}_1+\vec{V}_2+\vec{V}_3$
2. For the coplanar system of vectors $\vec{V}_1$ and $\vec{V}_2$ shown below



a) Determine the magnitude of vector sum $\vec{V}=\vec{V}_1+\vec{V}_2$ and the angle θ_x which $\vec{V}$ makes with the positive x-axis.

b) Write $\vec{V}$ as a vector in terms of the unit vectors $\hat{\imath}$ and $\hat{\jmath}$ and then write the unit vector along vector$\vec{V}$.

c) Determine the magnitude of vector difference $\vec{V}=\vec{V}_2-\vec{V}_1$ and angle θ_x which $\vec{V}$ makes with the positive x-axis.

d) Determine the magnitude of vector difference $\vec{V}=\vec{V}_1-\vec{V}_2$ and the angle θ_x which $\vec{V}$ makes with x-axis.

3. Find the value of k for which the vectors $\overrightarrow{A}$= 4$\hat{\imath}$-2$\hat{\jmath}$+8$\hat{k}$ and $\overrightarrow{B}$= 12$\hat{\imath}$-6$\hat{\jmath}$+k$\hat{k}$ are parallel.
4. The resultant of the two forces, when they act at angle of 60° is 14N. If the forces are acting at right angles, their resultant is $\sqrt{136}$N. Determine the magnitude of the two forces.

5. Three forces having a magnitude of 40KN, 15KN, & 20KN are acting at a certain point in the x,y plane. The angles made by 40KN, 15KN, & 20KN forces with the positive x-axis in CCW direction are 60°, 120°, and 240° respectively. Determine the magnitude and direction of the resultant force.

6. Determine the magnitude and direction of vector $\vec{V}_2$ such that $\vec{V}$= 2$\vec{V}_1$ + $\vec{V}_2$ acts along the positive z-axis having a magnitude of 20 from which vector $\vec{V}_1$ = 20$\hat{\imath}$-15$\hat{\jmath}$+15$\hat{k}$.